

ARTICLE

Optimized Fabrication and Characterization of W Tips for High-Performance Atomic-Resolution Scanning Tunneling Microscope

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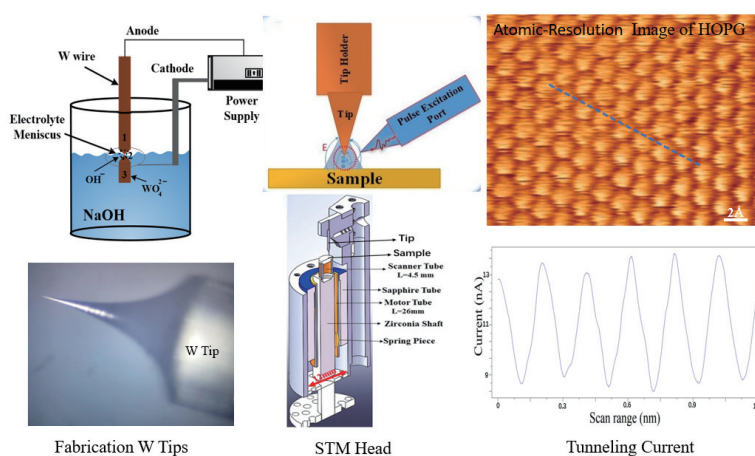
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Scanning tunneling microscope (STM) is a powerful technique for atomic-resolution surface characterization that relies on quantum tunneling through sharp conductive probes. The preparation of high-quality STM tips represents a critical aspect of instrument performance,

where tip sharpness determines imaging resolution and mechanical stability ensures measurement reproducibility. Tungsten (W) is widely used for STM tips due to its exceptional mechanical strength and electrical conductivity. In this study, we optimized the fabrication of W tips through electrochemical etching by systematically investigating etching time, wire immersion depth, and applied voltage. Multiple tips were fabricated and evaluated based on their curvature radius and sharpness. The imaging performance of these tips was assessed through atomic-resolution STM measurements on highly oriented pyrolytic graphite (HOPG) at room temperature using a home-built STM system. Among the fabricated tips, the sharpest specimens demonstrated outstanding performance under extreme conditions, including low temperatures and high magnetic fields, confirming their suitability for advanced applications. These findings establish a reliable framework for fabricating high-performance W tips, which are essential for precise STM imaging and can function effectively as antennas in terahertz (THz)-driven STM systems for exploring ultrafast nanoscale dynamics.

Key words: Scanning tunneling microscope, Electrochemical etching method, High-resolution imaging, High stability, Low temperature and High magnetic fields



I. INTRODUCTION

Scanning tunneling microscope (STM) is a powerful

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technique for investigating surface structures at the atomic scale, playing a vital role in condensed matter physics, nanotechnology, and materials science [1–3]. Since its invention, STM has allowed researchers to visualize electronic and structural properties of materials with unparalleled spatial resolution [4–7]. STM has also emerged as a vital technique for probing quantum materials under extreme conditions, such as low temperatures and high magnetic fields, enabling atomic-scale characterization of their electronic and magnetic structures. Among these materials, non-collinear anti-ferromagnetic half-metals with their unconventional magnetic ordering and unique band structures offer a compelling platform for STM investigations. Their complex topological features can be directly visualized at the atomic level, advancing both fundamental understanding and future device applications [8–10]. Similarly, two-dimensional (2D) magnetic materials, known for their atomically flat surfaces and intrinsic magnetic order, are particularly well-suited for STM studies in the presence of external magnetic fields. These materials facilitate a precise exploration of magnetic phenomena at the nanoscale, providing critical insights into spintronic functionalities and low-dimensional magnetism [11, 12]. Attaining high-resolution imaging under extreme experimental conditions essentially relies on the quality and reliability of the STM tip, positioning tip fabrication as a crucial aspect in the continued development and performance of STM-based investigations.

The resolution and quality of STM images are fundamentally dependent on the sharpness of the metallic tip, which plays a pivotal role in ensuring accurate measurements [13–15]. STM provides essential insights into the atomic-scale topography and electronic characteristics of conductive materials by measuring the tunneling current between the tip and the sample surface [16–18]. The sharpness of the tip apex directly enhances image quality, enabling precise atomic-scale imaging. Over the past few decades, various fabrication techniques have been developed to produce different metallic tips, with electrochemical etching [19–23] emerging as the most practical and reliable method for fabricating STM tips that meet the desired standards of quality and reproducibility.

Electrochemical etching is a commonly employed technique for fabricating tips from materials like tungsten, platinum, and platinum-iridium [24–26]. The tip must possess excellent electrical conductivity, robust

mechanical strength, and long-term durability for STM applications [27, 28]. These properties ensure that the tip resists deformation during imaging, resulting in high repeatability and reliability of the obtained images. Among various materials, tungsten is generally the preferred choice for STM tip fabrication due to its exceptional mechanical strength and good electrical conductivity [29, 30].

Although the fabrication of tungsten STM tips using electrochemical etching has been widely studied, several key factors, influencing the sharpness and quality of the tips, remain underexplored. Several factors influence the electrochemical etching process, including the type of applied voltage (AC or DC), the timing of voltage deactivation, the diameter of the tungsten wire, environmental stability, the composition of the electrolyte, and the alignment of the tungsten wire. However, there has been no detailed and systematic exploration of how these parameters affect the sharpness of the resulting tips in STM imaging.

The STM tip also plays a crucial role in the coupling, excitation, and enhancement of terahertz (THz) light in THz-driven STM [31]. The tip's role becomes even more significant in THz-STM, as it functions like an antenna [32–34]. The key mechanisms involved include dipole effects between the tip and the sample, plasmonic excitations, and resonant effects due to the antenna-like behavior of the tip in response to incident wavelengths. Additionally, the localized enhancement of free electrons near the metal nanostructure and the high-density charge concentration effects arising from the protruding tip contribute to the amplification of THz signals [34–38].

This study presents a systematic investigation into the electrochemical fabrication of W tips for STM, aimed to optimize their performance for both conventional and THz-driven STM applications. Key fabrication parameters, including etching temperature, duration, applied voltage, and immersion depth, were varied to enhance tip sharpness and reproducibility. The fabricated tips were evaluated on a highly oriented pyrolytic graphite (HOPG) surface to assess their atomic-resolution imaging capabilities. These tips were integrated into a home-built STM system capable of stable operation at room temperature and under extreme conditions, such as cryogenic temperatures and high magnetic fields. The STM head features a fully open structural design, allowing unobstructed optical access to the

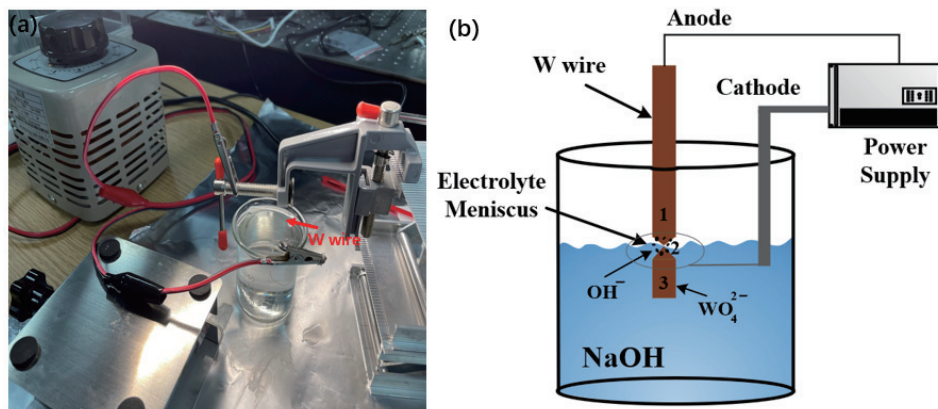


FIG. 1 (a) Photograph of the setup used for the preparation of STM tips. (b) Schematic illustration of the experimental setup for tungsten (W) tip fabrication via electrochemical etching.

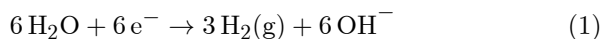
tip-sample junction and ensuring efficient THz field coupling for lightwave-driven STM applications. In this context, the tungsten tip plays a dual role as both a high-resolution imaging probe and an effective THz antenna, highlighting its critical importance for investigating ultrafast dynamics and quantum phenomena at the nanoscale.

II. EXPERIMENTAL PROCEDURE

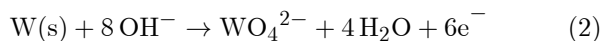
The fabrication of W tips was conducted using the electrochemical etching method. A W wire was vertically immersed in a sodium hydroxide (NaOH) solution within a cathode tube. FIG. 1 (a) and FIG. 1 (b) show a photograph of the setup and a schematic representation of the experimental configuration, respectively. The tungsten wire served as the anode, while the surrounding tube formed the cathode. A voltage was applied across these electrodes to initiate the etching process.

The reactions governing the electrochemical etching process are as follows [39]:

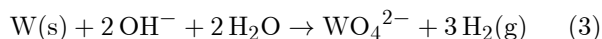
Cathode reaction:



Anode reaction:



Overall reaction:



Upon immersing the tungsten wire into the electrolyte, a meniscus develops at the neck of the wire,

defining the etching region, as illustrated in FIG. 1(b). This meniscus triggers a radial flow of hydroxide (OH^-) ions from the cathode to the anode. The soluble tungstate ions (WO_4^{2-}) generated during this reaction are carried downward by the OH^- flow, forming a dense, viscous layer around the lower end of the wire. The higher concentration of OH^- ions at the base of the meniscus (region 2) results in a faster etching rate compared to the upper region of the meniscus (region 1). The dense WO_4^{2-} layer that accumulates around the lower end of the wire (region 3) inhibits further etching in that area, causing a necking effect in region 2. As the wire narrows at this point, its tensile strength diminishes until it can no longer support the weight of the lower portion, leading to its breakage. This process forms a sharp tip with a curvature radius of 20–100 nm.

The sharpness of the nanotip apex is highly sensitive to the timing of voltage deactivation. Failure to promptly turn off the applied voltage after the wire breaks can lead to continued etching at the tip apex, resulting in an increased radius and a loss of sharpness. Therefore, precisely switching off the voltage is critical for achieving ultra-sharp nanotip apexes.

III. STM HEAD

The STM head consists of several carefully engineered components to ensure optimal performance. The main structural element is a tantalum frame, chosen for its mechanical stability and resistance to thermal expansion, with dimensions of 20 mm outer diameter (OD) and 51 mm length. We designed this scanning unit for our future research on THz-driven STM, aiming to achieve precision, stability, and high-resolution

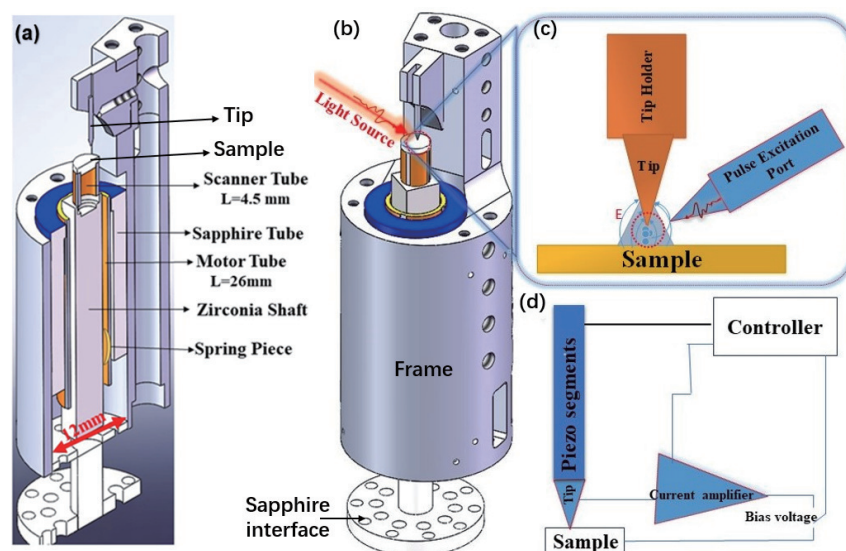


FIG. 2 (a) Cross-sectional view of the STM head. (b) Three-dimensional model of the assembled STM head. (c) Schematic illustration of the lightwave-driven STM mechanism. (d) Experimental setup featuring a feedback loop for controlling the tip-sample distance.

imaging in nanoscale investigations. The working principle and application of this single-arm STM in a dry magnet were detailed in our previous study [18]. The system incorporates two piezoelectric tubes (PTs), which are essential for both coarse positioning and high-precision scanning. The motor tube (7.09 mm OD, 26 mm length) is responsible for the coarse movement of the tip, allowing large-scale adjustments, while the scanning tube (3.5 mm OD, 4.5 mm length) provides nanometer-scale precision for fine imaging. Other essential components include a tip holder and sample holder, ensuring stable tip-sample interaction, a zirconia shaft (27 mm length) that enhances mechanical rigidity, a sapphire tube, and a spring piece for additional structural support and flexibility. These elements work together to optimize the stability and accuracy of the STM operation. The mechanical structure and assembly of the STM head are illustrated in FIG. 2 (a) and (b), which present a section view and a three-dimensional model, respectively. These visualizations provide a comprehensive overview of the component layout and design considerations that contribute to the system's stability and compatibility with extreme experimental conditions.

The STM head incorporates a fully open structural design, which is particularly beneficial for lightwave-driven STM applications as it allows for unobstructed optical access to the tunneling junction. The system operates based on a lightwave-driven mechanism, where a single-cycle THz pulse is directed onto the STM tip, as

shown in FIG. 2(b). This setup enables ultrafast modulation of the tunneling junction, opening new possibilities for high-frequency spectroscopic investigations. The tungsten tip functions as a broadband optical antenna, efficiently coupling the THz electric field into the junction by exciting a Sommerfeld mode [40], as illustrated in FIG. 2(c). This mode, confined to the surface of the tip, enhances the propagation and confinement of the electromagnetic fields along the tungsten shaft, significantly improving the temporal resolution of the STM.

The W tip, mounted on a piezoelectric positioning system, is precisely controlled with sub-nanometer accuracy to approach a conductive sample. When a bias voltage is applied, quantum tunneling occurs, allowing electrons to transfer between the tip and the sample. This tunneling current is then detected by a low-noise current amplifier, which serves as the primary signal for both topographic imaging and local spectroscopic measurements. A real-time feedback loop, as shown in FIG. 2(d), dynamically adjusts the tip-sample distance to maintain a constant tunneling current. This feedback system is critical for maintaining stable imaging resolution and compensating for thermal drift and external mechanical disturbances, ensuring the precision and reliability of the STM measurements.

IV. EXPERIMENTAL RESULTS

The experimental conditions for W tip fabrication were carefully monitored, with ambient laboratory tem-

TABLE I Electrochemical etching parameters for fabrication of tungsten tips.

Tip number	Electrolyte concentration/(mol/L)	Applied voltage/V	Length/mm	Ething time/s
1	3.5	10	7	613±10
2	3.5	9	8	623±10
3	3.5	7.5	7	553±10
4	3.5	8	10	430±10

peratures recorded throughout the process, ranging from 19 °C to 28 °C. These temperature measurements, taken during active experimentation, revealed significant variations in etching behavior. Key experimental parameters, including the fixed electrolyte concentration (3.5 mol/L NaOH/KOH solution), applied DC voltage (7.5–10 V), and tungsten wire immersion length (7–10 mm), were systematically varied to thoroughly examine their collective influence on both etching kinetics and final tip morphology. The complete matrix of these controlled conditions and their corresponding etching durations are presented in Table I.

At the lowest recorded ambient temperature (~19 °C, corresponding to tip 1), the electrochemical etching process proceeded markedly slowly, resulting in an extended etching duration of 613 s. This reduced reaction rate, attributable to decreased ionic mobility at lower temperatures, not only prolonged the fabrication process but also increased the probability of meniscus rupture events. Consequently, under these conditions, tips were frequently yielded with less-defined neck regions and inconsistent taper profiles, as shown in FIG. 3(a). When the temperature was slightly elevated to ~22 °C (tip 2 conditions), the etching process demonstrated improved efficiency, completing in 623 s while producing tips with more desirable geometric characteristics and superior surface smoothness, as revealed in FIG. 3(b).

The most favorable results were obtained under conditions corresponding to tip 3, fabricated at ~25 °C with an applied potential of 7.5 V and an immersion depth of 7 mm. This optimal parameter combination consistently produced tips exhibiting an ideal exponential taper profile with an average apex curvature radius of ~28 nm, as verified by subsequent SEM characterization, as shown in FIG. 3(c). Such tips demonstrated an exceptional balance between atomic-scale sharpness and mechanical stability. In stark contrast, the most aggressive etching conditions (~28 °C, tip 4) produced the

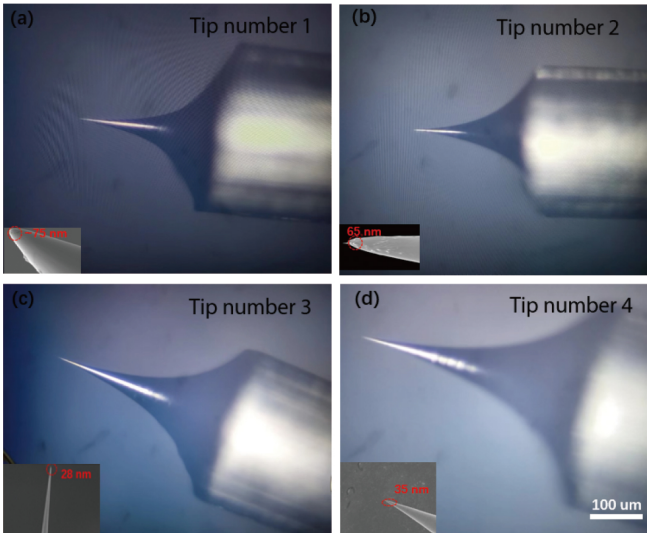


FIG. 3 Optical images of (a) tip 1; (b) tip 2; (c) tip 3, exhibiting optimal sharpness and curvature; and (d) tip 4.

shortest etching time (430 s), but this came at the expense of process control. The accelerated reaction rate at elevated temperatures frequently induced meniscus instabilities, leading to increased tip asymmetry and occasional fracture at the critical necking region, as illustrated in FIG. 3(d).

These comprehensive results demonstrate that moderate ambient temperatures (~25 °C) provide the ideal conditions for controlled reproducible electrochemical etching. Under these conditions, the process yields high-quality tungsten tips with both the sharpness required for atomic-resolution STM imaging and the structural integrity needed for demanding applications, including THz-STM experiments. The temperature dependence observed in this study provides crucial insights for optimizing tip fabrication protocols across varying laboratory environments.

The performance of the fabricated W tips was evaluated using highly oriented pyrolytic graphite (HOPG) samples by employing a home-built STM system. These tests were conducted to assess the stability and imaging quality of these tips under various experimental conditions. HOPG was selected as the reference material due to its atomically flat surface and well-defined layered structure, which enables precise evaluation of the tip’s resolution and tunneling characteristics. FIG. 4 presents STM images of the HOPG surface obtained at room temperature using the fabricated tips. Specifically, FIG. 4 shows STM images captured with tip 1, tip 2, tip 3, and tip 4, respectively.

During the evaluation, the STM system successfully

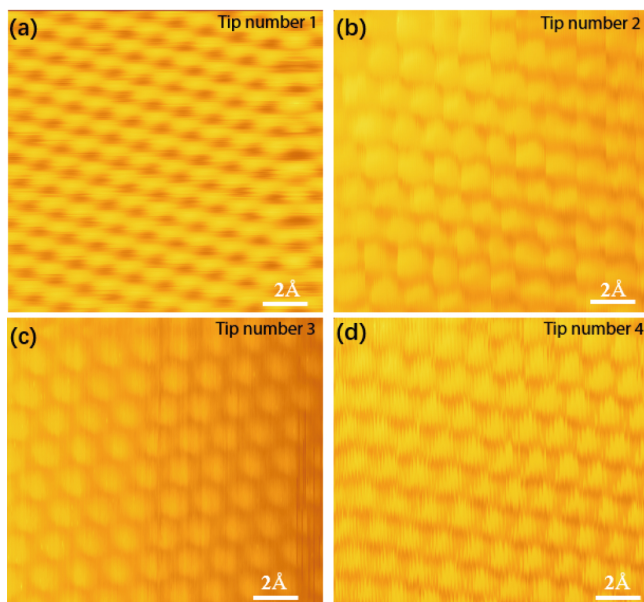


FIG. 4 Atomic-resolution STM images (raw data) of the HOPG surface obtained using different tungsten (W) tips under constant-height mode. (a) Image acquired with tip 1. (b) Image acquired with tip 2. (c) Image acquired with tip 3. (d) Image acquired with tip 4. Imaging parameters: temperature of 300 K, tunneling current of 10 nA, bias voltage of +100 mV.

achieved atomic-resolution imaging, demonstrating the effectiveness of the fabricated tungsten tips. The observed stability in the tunneling current confirmed the high quality of the tip apex, which is crucial for minimizing tip-induced artifacts during imaging. Moreover, the ability to control the tip-sample distance with sub-nanometer precision underscores the effectiveness of the piezoelectric control system and the real-time feedback loop.

These evaluations validate the STM design and provide a solid foundation for its application in future THz-STM research. Such research will enable the exploration of ultrafast electronic and quantum phenomena at the atomic scale. The integration of lightwave-driven tunneling, precise piezoelectric control, and high-resolution imaging establishes this STM system as a powerful tool for investigating ultrafast charge dynamics, molecular interactions, and surface electronic states in nanomaterials.

The relationship between applied voltage and etching time is presented in FIG. 5(a). The results reveal a distinct trend where the etching time decreases with increasing voltage up to 8.5 V, with the shortest etching time being recorded at 8.5 V (540 s). Beyond this point, the etching time begins to rise, reaching 620 s at 9 V.

This trend can be attributed to the interplay between the etching rate and the stability of the electrochemical process. At lower voltages (4–6 V), the etching process progresses slowly due to insufficient driving force for ionization, resulting in longer etching times. Conversely, at higher voltages (8–9 V), the etching process becomes more aggressive but less controlled, potentially introducing irregularities in the tip geometry. This emphasizes that 8.5 V represents an optimal condition for achieving efficient and controlled etching.

The average curvature radius of the fabricated tips was evaluated as a function of applied voltage, as shown in FIG. 5(b). Tips produced at lower voltages (4–6 V) exhibited curvature radii ranging from 30 nm to 45 nm. The smallest curvature radius, 28 nm, was achieved at 7 V. However, increasing the voltage beyond this point led to a significant increase in the curvature radius, with tips fabricated at 9 V exhibiting an average curvature radius of 60 nm.

These results demonstrate that intermediate voltages (6–8 V) reach the best balance etching speed and tip sharpness. The smallest radius at 7 V is particularly noteworthy, as it indicates the potential for high-resolution STM imaging. The increase in curvature radius at higher voltages can be attributed to excessive material removal, which compromises the tip's sharpness.

To evaluate the robustness and performance of the STM head and the fabricated W tips under extreme and confined environments, tip 3 was tested in a low-temperature setup with a high magnetic field. Experiments were conducted using a cryogen-free superconducting magnet (model: TeslatronPT, Oxford Instruments), enabling measurements at cryogenic temperatures and in strong magnetic fields. FIG. 6(a) presents an STM image of the HOPG surface acquired in constant-height mode at 1.8 K and under a magnetic field of up to 10 T. These conditions are critical for applications involving quantum materials and ultrafast dynamics, where both precise imaging and mechanical stability are essential. The image demonstrates the capability of tip 3 to maintain stability and resolve atomic-scale features even under such demanding experimental conditions.

FIG. 6(b) shows the corresponding tunneling current profile measured along the dashed line in FIG. 6(a). The profile highlights the uniformity of the tunneling current, further confirming the sharpness and reliability of tip 3. These results validate that the fabricated tungsten tips can operate effectively in confined, low-

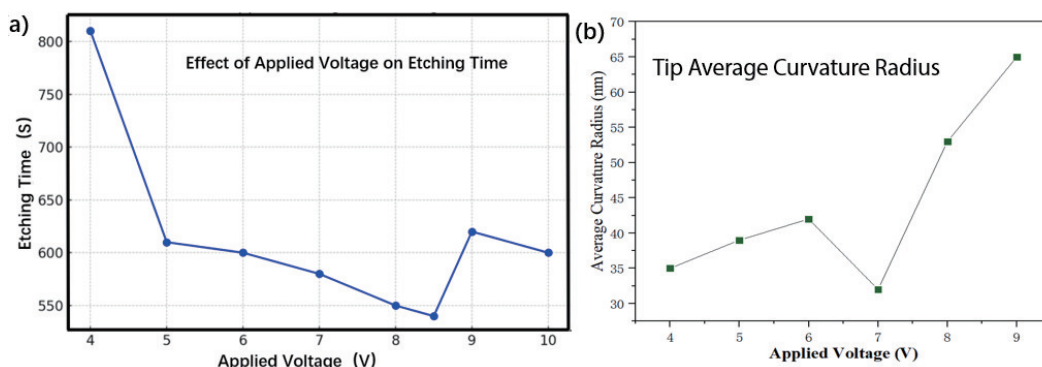


FIG. 5 (a) Effect of the applied voltage on etching time. (b) Average curvature radius of the tip as a function of applied voltage.

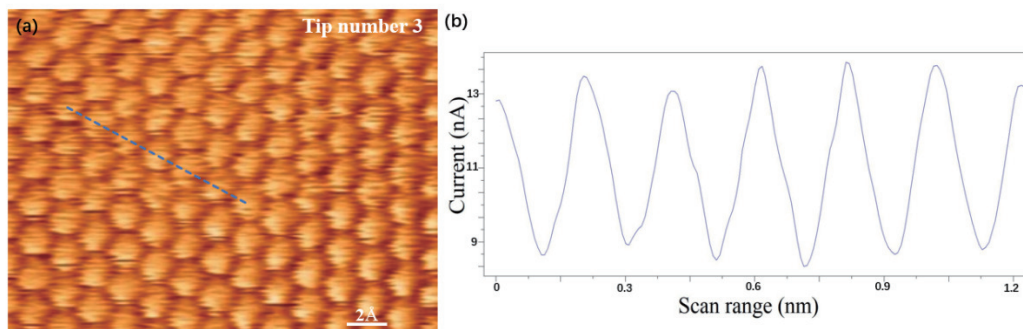


FIG. 6 Atomic-resolution STM imaging of HOPG acquired in constant-height mode under extreme conditions. (a) Raw STM image showing the atomic lattice, obtained using tip 3 at a magnetic field of 10 T. (b) Corresponding tunneling current profile along the dashed line in (a), revealing periodic atomic features. Measurement parameters: temperature of 1.8 K, tunneling current of 10 nA, sample bias of +100 mV.

temperature, and high-magnetic-field environments, making them suitable for advanced investigations in condensed matter physics and related fields.

We systematically evaluated the stability of our STM system equipped with tip 3 to evaluate its long-term performance and reliability under continuous scanning conditions. This tip was selected based on its superior sharpness and consistent performance observed during initial characterization tests. The exceptional sharpness of tip 3 is crucial for achieving high-resolution imaging, while its stability ensures reliable operation over extended periods, essential for applications requiring precise, high-quality measurements.

To quantitatively assess system stability, we performed atomic-resolution imaging of the same graphite surface area continuously for 3 h. The system demonstrated excellent positional stability, with a minimal displacement of just 4.6 nm for reference markers, as shown in FIG. 7(a–c). Additional stability metrics included measured drift rates of 21.4 pm/min in the X - Y plane (FIG. 7(d)) and 25.9 pm/min along the Z -axis, as shown in FIG. 7(e).

These remarkably low drift rates at 300 K are particularly advantageous for STM applications across physics, chemistry, materials science, and biological research, where maintaining stability during prolonged measurements is critical for obtaining reliable data. The outstanding performance of tip 3 confirms its suitability for advanced STM applications requiring both high resolution and long-term stability, including studies of ultrafast dynamics and nanoscale phenomena.

The performance of STM is heavily dependent on the material used for the STM tip, which directly influences its sharpness, stability, and overall performance in high-resolution imaging. Various materials have been utilized for STM tips, each offering distinct advantages and limitations based on factors such as sharpness, durability, conductivity, and ease of fabrication. For instance, tungsten is widely used due to its high sharpness and conductivity, which make it particularly suitable for high-resolution imaging, even under harsh experimental conditions. However, its susceptibility to oxidation under high-temperature or high-vacuum conditions limits its long-term stability. Other materials,

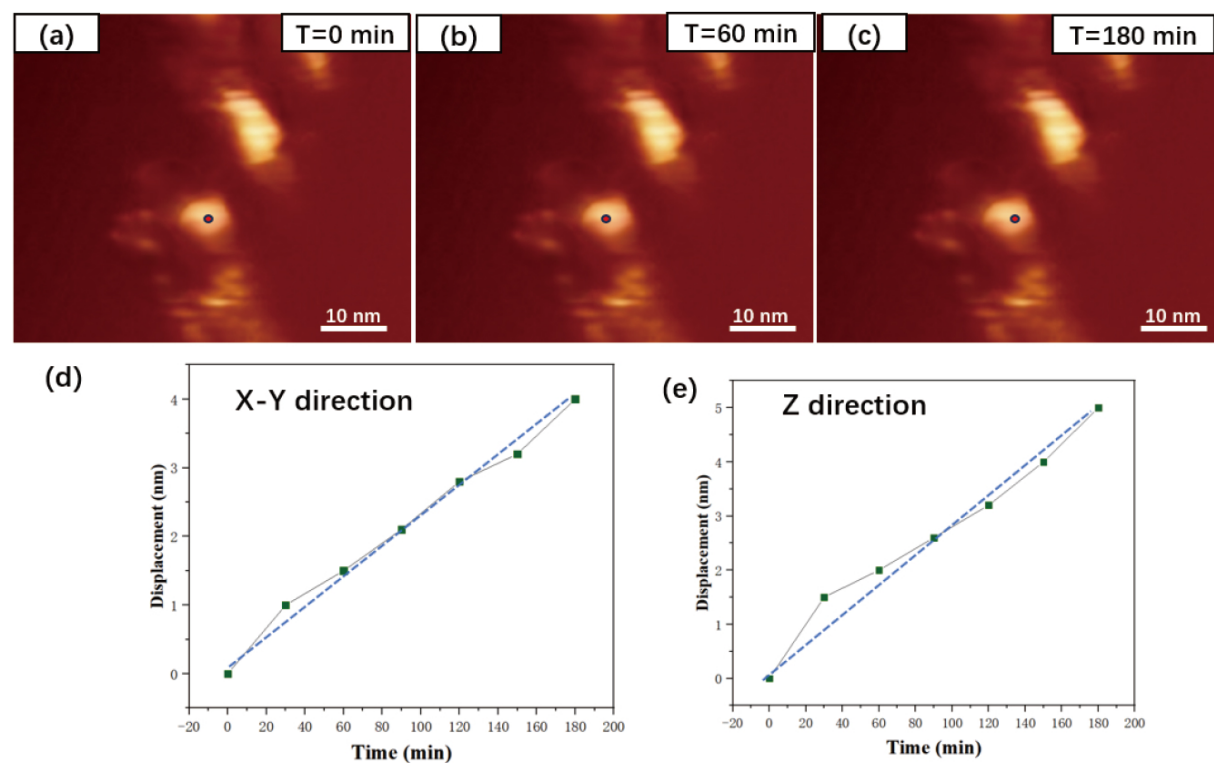


FIG. 7 Drift analysis using tip 3 over 3 h under constant-current mode. (a–c) Sequential raw STM images of the graphite surface. Imaging parameters: tunneling current of 7.5 nA, scan speed of 1 line/s, and bias voltage of +100 mV. (d) Drift rate in the X–Y plane. (e) Drift rate in the Z direction, with the tunneling current set at 7.5 nA. All measurements were performed at 300 K. Blue dashed lines indicate linear fits.

TABLE II Comparison of materials used for STM tip fabrication.

Material	Sharpness	Durability	Conductivity	Advantages	Limitations
W	Very high	Moderate to high	High	High sharpness, easy to fabricate, good electrical conductivity	Susceptible to oxidation
Pt	Moderate	High	High	Stable, resistant to oxidation, reliable for long-term use	Lower sharpness than W, more expensive
Pt-Ir	High	Very high	High	Extremely durable, resistant to oxidation	More expensive, difficult to fabricate sharp tips
Au	Moderate to high	Low to moderate	High	Easy to fabricate, excellent conductivity	Soft, easily deformed, low resolution

such as platinum and platinum–iridium, are known for their durability and resistance to oxidation, although they typically offer lower sharpness compared to tungsten.

To provide a clearer understanding of the relative performance of different materials, we present a comparison of commonly used STM tip materials, including tungsten, platinum, and platinum–iridium. This comparison highlights key properties such as sharpness, durability, and conductivity, which are essential for high-performance STM operations. The information provided in Table II serves to guide the selection of the most suitable material based on the specific require-

ments of the experiment, whether it is high-resolution imaging, long-term durability, or operating under extreme conditions.

To evaluate the performance of different tip materials under extreme conditions, we conducted STM imaging at 1.8 K and 10 T using both a platinum–iridium (Pt–Ir) tip and a W tip (tip 4). FIG. 8(a) presents an STM image acquired using a Pt–Ir tip (with a composition of 90% Pt and 10% Ir, annealed; sourced from Goodfellow Cambridge Limited, UK). The tip was prepared by mechanically cutting the wire at an angle ranging from 15° to 30°, a common method that enables adequate sharpness for surface imaging.

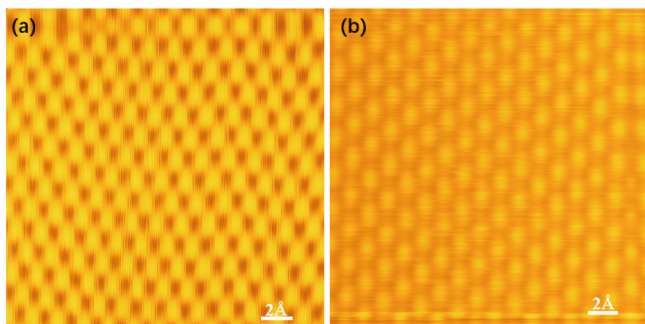


FIG. 8 Atomic-resolution STM images of graphite acquired in constant-height mode using different tip materials. (a) Image obtained with a Pt-Ir tip. (b) Image obtained with a W tip. Measurement parameters: bias voltage of +100 mV; tunneling current of 10 nA.

In contrast, FIG. 8(b) shows an STM image captured with our electrochemically etched W tip (tip 4) under low-temperature and high-magnetic-field conditions. Both sets of measurements were carried out at the Oxford Instruments Shanghai demonstration facility using a cryogen-free superconducting magnet system (TeslatronPT, Oxford Instruments Nanotechnology Tools Ltd., UK). This side-by-side comparison highlights the relative performance of the two tip types under harsh experimental conditions. While the Pt-Ir tip yielded stable imaging, the W tip demonstrated superior resolution and contrast at the atomic level, consistent with its higher sharpness as achieved through optimized etching. These results further validate the effectiveness of our tungsten tip preparation method for advanced STM applications in extreme environments.

V. DISCUSSION

The fabrication of tungsten tips with apex radii as small as 28 nm was successfully achieved through optimized direct current (DC) electrochemical etching. These tips demonstrated excellent performance in STM under extreme and confined conditions, providing atomic-scale resolution. However, consistently producing ultra-sharp tips remains a challenging task due to various factors that influence the etching process.

The etching process is susceptible to fluctuations in conditions such as voltage stability, electrolyte composition, and environmental factors. Additionally, impurities in the tungsten wires contribute to variability in both tip sharpness and durability. While DC etching is a straightforward and cost-effective technique, it faces limitations in consistently producing tips with radii smaller than 30 nm. Achieving reproducibility at sub-

nanometer scales remains a significant challenge. Although commercial tips with radii below 10 nm provide superior resolution, they are prohibitively expensive and fragile, highlighting the need for more robust and cost-efficient fabrication methods.

To address these challenges, several key parameters were optimized. The applied voltage was found to play a crucial role in both the etching time and tip sharpness. An intermediate voltage of 7 V yielded the sharpest tips, with an apex radius of approximately 28 nm. Lower voltages resulted in slower etching due to insufficient ionization, while higher voltages accelerated the process but led to irregular geometries and compromised stability.

The electrolyte composition was also critical, with a NaOH-based solution at a concentration of 3.5 mol/L producing the most stable and consistent etching results. Environmental factors such as vibrations and deviations from vertical alignment further disrupted the uniformity of the etching process. Ensuring precise vertical alignment of the tungsten wire and isolating the setup from external vibrations significantly improved tip consistency. Moreover, a tungsten wire diameter of 7 mm was found to be optimal for producing sharp and reproducible tips, as larger diameters led to slower etching and inconsistent results.

To verify the performance of the fabricated tungsten tips, they were tested on HOPG samples using a home-built STM. Tips with radii around 28 nm consistently resolved atomic-scale features with high contrast, demonstrating their suitability for high-resolution imaging. In contrast, tips with larger radii produced lower-quality images with reduced resolution and contrast, emphasizing the critical role of tip sharpness in STM performance.

Despite these advancements, further refinements are necessary to produce ultra-sharp tips with radii smaller than 10 nm, which would enable even higher resolution for both STM and THz-STM applications. Exploring advanced etching techniques, such as pulse etching or the use of alternative electrolytes, could improve tip sharpness and reproducibility. Additionally, automation through real-time monitoring and feedback systems would provide precise control over etching parameters. The use of purer tungsten wires and protective coatings could enhance the durability of the tips. Integrating these optimized tips into THz-STM systems will offer new opportunities to study ultrafast nanoscale dy-

namics and optoelectronic properties. By addressing the identified challenges and leveraging advanced fabrication techniques, the goal of achieving ultra-sharp tips with radii below 10 nm becomes achievable, paving the way for further advances in STM and THz-STM research.

VI. CONCLUSION

This study successfully optimized the electrochemical etching process for fabricating tungsten (W) tips with apex radii as small as 28 nm, suitable for achieving high atomic-resolution STM imaging. Key factors such as applied voltage, electrolyte concentration, tungsten wire alignment, and environmental isolation were identified to be critical for ensuring consistent tip sharpness and reproducibility. Optimal etching conditions were established using a 3.5 mol/L NaOH electrolyte and an applied voltage of 7.5 V, resulting in high-quality tip formation. To evaluate their performance, the fabricated tips were integrated into a custom-built STM system with an open-access and single-arm structure. Imaging of graphite demonstrated that the W tips consistently resolved atomic-scale features, confirming their effectiveness for high-resolution surface characterization. This STM system is specifically designed to be highly suitable for THz-driven applications, enabling precise imaging and optimal sensitivity for investigating ultrafast dynamics and quantum phenomena. The W tips also function efficiently as THz antennas, significantly advancing research in nanostructures, quantum materials, and ultrafast dynamics in lightwave-driven STM systems [41–44], as well as optoelectronic phenomena in THz-STM platforms [45–47].

VII. ACKNOWLEDGMENTS

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- [1] B. Jäck, Y. Xie, J. Li, S. Jeon, B. A. Bernevig, and A. Yazdani, *Science* **364**, 1255 (2019).
- [2] S. Y. Li, Z. W. Wang, Y. C. Xue, Y. B. Wang, S. H. Zhang, J. P. Liu, Z. Zhu, K. Watanabe, T. Taniguchi, H. J. Gao, Y. H. Jiang, and J. H. Mao, *Nat. Commun.* **13**, 4225 (2022).
- [3] W. Meng, Y. Guo, Y. Hou, and Q. Lu, *Nano Res.* **8**, 3898 (2015).
- [4] J. Wang and M. Sun, *Micron* **187**, 103719 (2024).
- [5] N. Jiang, E. T. Foley, J. M. Klingsporn, M. D. Sonntag, N. A. Valley, J. A. Dieringer, T. Seideman, G. C. Schatz, M. C. Hersam, and R. P. Van Duyne, *Nano Lett.* **12**, 5061 (2012).
- [6] J. Liu, K. Zhao, A. Wang, M. Zhang, J. Wang, Y. Hou, W. Meng, and Q. Lu, *Chin. J. Chem. Phys.* (2024). DOI: [10.1063/1674-0068/cjcp2405077](https://doi.org/10.1063/1674-0068/cjcp2405077).
- [7] M. Fernández-Lomana, B. Wu, F. Martín-Vega, R. Sánchez-Barquilla, R. Álvarez-Montoya, J. M. Castilla, J. Navarrete, J. R. Marijuan, E. Herrera, H. Suderow, and I. Guillamón, *Rev. Sci. Instrum.* **92**, 093701 (2021).
- [8] Y. Deng, X. Liu, Y. Chen, Z. Du, N. Jiang, C. Shen, E. Zhang, H. Zheng, H. Z. Lu, and K. Wang, *Natl. Sci. Rev.* **10**, 154 (2023).
- [9] X. Liu, Q. Feng, D. Zhang, Y. Deng, S. Dong, E. Zhang, W. Li, Q. Lu, K. Chang, and K. Wang, *Adv. Mater.* **35**, 2211634 (2023).
- [10] L. Min, H. Tan, Z. Xie, L. Miao, R. Zhang, S. H. Lee, V. Gopalan, C. X. Liu, N. Alem, B. Yan, and Z. Mao, *Nat. Commun.* **14**, 364 (2023).
- [11] S. Zhang, J. Wu, M. Zheng, X. Jin, Z. Shen, Z. Li, Y. Wang, Q. Wang, X. Wang, H. Wei, J. Zhang, P. Wang, S. Zhang, L. Yu, L. Dong, Q. Zhu, H. Zhang, and J. Lu, *Nat. Commun.* **14**, 3634 (2023).
- [12] W. Zhu, H. Lin, F. Yan, C. Hu, Z. Wang, L. Zhao, Y. Deng, Z. R. Kudrynskyi, T. Zhou, Z. D. Kovalyuk, Y. Zheng, A. Patané, I. Žutić, S. Li, H. Zheng, and K. Wang, *Adv. Mater.* **33**, 2104658 (2021).
- [13] A. J. Melmed, *J. Vac. Sci. Technol. B* **9**, 601 (1991).
- [14] M. G. Boyle, L. Feng, and P. Dawson, *Ultramicroscopy* **108**, 558 (2008).
- [15] B. Li, Y. Zhang, J. Wang, Z. Jia, C. Shi, Y. Ma, and L. Ma, *SN Appl. Sci.* **2**, 1246 (2020).
- [16] M. Touqeer, B. Esmaeilzadeh, W. Meng, J. Wang, S. A. Maqbool, S. Zheng, L. Junwei, Y. Hou, and Q. Lu, *Ultramicroscopy* **263**, 113983 (2024).
- [17] Y. Yang and Y. Xu, *J. Microsc.* **292**, 37 (2023).
- [18] B. Esmaeilzadeh, M. Touqeer, L. Junwei, S. Zheng, T. Geng, Y. Hou, and Q. Lu, *J. Microsc.* **294**, 26 (2024).
- [19] G. Tahmasebipour, V. Ahmadi, A. Abdullah, and Y. Hojjat, *Int. J. Nanoscience* **8**, 305 (2009).
- [20] J. P. Ibe, P. P. Bey, S. L. Brandow, R. A. Brizzolara, N. A. Burnham, D. P. DiLella, K. P. Lee, C. R. K.

- Marrian, and R. J. Colton, *J. Vac. Sci. Technol. A* **8**, 3570 (1990).
- [21] A. B. Milhim and R. Ben Mrad, *J. Vac. Sci. Technol. B* **32**, 031806 (2014).
- [22] Y. Khan, H. Al-Falih, Y. Zhang, T. K. Ng, and B. S. Ooi, *Rev. Sci. Instrum.* **83**, 063708 (2012).
- [23] J. Zhang, P. Wang, X. Zhang, H. Ji, J. Luo, H. Wang, and J. Wang, *Rev. Sci. Instrum.* **92**, 015124 (2021).
- [24] P. Kim, J. H. Kim, M. S. Jeong, D. K. Ko, J. Lee, and S. Jeong, *Rev. Sci. Instrum.* **77**, 063708 (2006).
- [25] B. Ren, G. Picardi, and B. Pettinger, *Rev. Sci. Instrum.* **75**, 837 (2004).
- [26] A. N. Chaika, N. N. Orlova, V. N. Semenov, E. Yu. Postnova, S. A. Krasnikov, M. G. Lazarev, S. V. Chekmazov, V. Yu. Aristov, V. G. Glebovsky, S. I. Bozhko, and I. V. Shvets, *Sci. Rep.* **4**, 3742 (2014).
- [27] I. Altfeder, H. Lee, J. Hu, R. D. Naguy, A. Sehirlioglu, A. N. Reed, A. A. Voevodin, and C. B. Eom, *Phys. Rev. B* **93**, 115437 (2016).
- [28] G. Tian and Y. Luo, *Angew. Chem. Int. Ed.* **52**, 4814 (2013).
- [29] I. Ekvall, E. Wahlström, D. Claesson, H. Olin, and E. Olsson, *Meas. Sci. Technol.* **10**, 11 (1999).
- [30] L. Ottaviano, L. Lozzi, and S. Santucci, *Rev. Sci. Instrum.* **74**, 3368 (2003).
- [31] T. L. Cocker, V. Jelic, M. Gupta, S. J. Molesky, J. A. J. Burgess, G. D. L. Reyes, L. V. Titova, Y. Y. Tsui, M. R. Freeman, and F. A. Hegmann, *Nat. Photonics* **7**, 620 (2013).
- [32] S. E. Ammerman, V. Jelic, Y. Wei, V. N. Breslin, M. Hassan, N. Everett, S. Lee, Q. Sun, C. A. Pignedoli, P. Ruffieux, R. Fasel, and T. L. Cocker, *Nat. Commun.* **12**, 6794 (2021).
- [33] J. Allerbeck, J. Kutteruff, L. Bobzien, L. Huberich, M. Tsarev, and B. Schuler, *ACS Photonics* **10**, 3888 (2023).
- [34] H. Azazoglu, P. Kapitza, M. Mittendorff, R. Möller, and M. Gruber, *Rev. Sci. Instrum.* **95**, 023703 (2024).
- [35] M. Abdo, S. Sheng, S. Rolf-Pissarczyk, L. Arnhold, J. A. J. Burgess, M. Isobe, L. Malavolti, and S. Loth, *ACS Photonics* **8**, 702 (2021).
- [36] Y. Zhang, C. Wang, B. Huai, S. Wang, Y. Zhang, D. Wang, L. Rong, and Y. Zheng, *Appl. Sci.* **11**, 71 (2021).
- [37] L. Bobzien, J. Allerbeck, S. E. Ammerman, R. Torsi, J. A. Robinson, and B. Schuler, *APL Mater.* **12**, 051110 (2024).
- [38] B. Esmaeilzadeh, M. Moradi, and F. Jahantigh, *J. Magn. Magn. Mater.* **460**, 207 (2018).
- [39] T. K. Yamada, T. Abe, N. M. K. Nazriq, and T. Iri-sawa, *Rev. Sci. Instrum.* **87**, 033703 (2016).
- [40] M. Müller, N. Martín Sabanés, T. Kampfrath, and M. Wolf, *ACS Photonics* **7**, 2046 (2020).
- [41] T. L. Cocker, D. Peller, P. Yu, J. Repp, and R. Huber, *Nature* **539**, 263 (2016).
- [42] S. Yoshida, Y. Arashida, H. Hirori, T. Tachizaki, A. Taninaka, H. Ueno, O. Takeuchi, and H. Shigekawa, *ACS Photonics* **8**, 315 (2021).
- [43] V. Jelic, K. Iwaszczuk, P. H. Nguyen, C. Rathje, G. J. Hornig, H. M. Sharum, J. R. Hoffman, M. R. Freeman, and F. A. Hegmann, *Nat. Phys.* **13**, 591 (2017).
- [44] C. Roelcke, L. Z. Kastner, M. Graml, A. Biereder, J. Wilhelm, J. Repp, R. Huber, and Y. A. Gerasimenko, *Nat. Photonics* **18**, 595 (2024).
- [45] T. L. Cocker, V. Jelic, R. Hillenbrand, and F. A. Hegmann, *Nat. Photonics* **15**, 558 (2021).
- [46] Y. Luo, V. Jelic, G. Chen, P. H. Nguyen, Y. J. R. Liu, J. A. M. Calzada, D. J. Mildenerger, and F. A. Hegmann, *Phys. Rev. B* **102**, 205417 (2020).
- [47] X. Wei, F. Yan, Q. Lv, C. Shen, and K. Wang, *Nanoscale* **9**, 8388 (2017).